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Inventor(s)	Masuda; Kazuki

Composite component

Abstract

A composite component includes a primary winding, a secondary winding, a coil, a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, and wiring. The primary winding and the secondary winding in combination function as a transformer. The secondary winding is connected between the third terminal and the fourth terminal. The fifth terminal functions as a center tap of the secondary winding. The sixth terminal is connected to the first terminal via the wiring and the primary winding. The sixth terminal is connected to the second terminal via the wiring and the coil. A first portion of the primary winding that is connected to the wiring, a second portion of the coil that is connected to the wiring, and the wiring constitute a three-way junction.

Inventors:	Masuda; Kazuki (Yokkaichi, JP)
Applicant:	AutoNetworks Technologies, Ltd. (Yokkaichi, JP); Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP); Sumitomo Electric Industries, Ltd. (Osaka, JP)
Family ID:	74570632
Assignee:	AutoNetworks Technologies, Ltd. (Yokkaichi, JP); Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP); Sumitomo Electric Industries, Ltd. (Osaka, JP)
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Primary Examiner: Lian; Mang Tin Bik

Assistant Examiner: Lian; Mangtin

Attorney, Agent or Firm: Honigman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national stage of PCT/JP2020/029619 filed on Aug. 3, 2020, which claims priority of Japanese Patent Application No. JP 2019-146946 filed on Aug. 9, 2019, the contents of which are incorporated herein.

TECHNICAL FIELD

(2) The present disclosure relates to a composite component.

BACKGROUND

(3) A phase-shift, full-bridge DC/DC converter is known. For example, a pair of clamp diodes connected in series between a high-voltage line and a low-voltage line on the primary side are adopted in JP 2017-127051A. Furthermore, two pairs of switch elements connected in series between the high-voltage line and the low-voltage line on the primary side are adopted.

Furthermore, an inductor connected between a connection point between each pair of switch elements and a connection point between the pair of clamp diodes is adopted.

(4) A device including an inductor and a transformer is known. For example, in JP 2011-82205A, a wound conductor portion of a primary coil of the transformer and a wound conductor portion of a secondary coil of the transformer are wound around a first core. The wound conductor portion of the secondary coil includes a first wound conductive wire and a second wound conductive wire. The inductor includes extended conductor portions respectively formed by extending the first wound conductive wire and the second wound conductive wire. The center tap of the inductor is connected to the secondary coil.

(5) In a phase-shift, full-bridge DC/DC converter (hereinafter tentatively referred to as a “full-bridge converter”), one end of an inductor and one end of a primary coil of a transformer are both connected to a connection point (hereinafter tentatively referred to as a “diode connection point”) between a pair of clamp diodes. To achieve this connection, it is preferable that the winding of the inductor is provided separately from the winding of the transformer. In this respect, it is not appropriate to directly use the inductor disclosed in JP 2011-82205A as the inductor of the full-bridge converter.

(6) When conductive wires that are respectively drawn out from the inductor and the primary coil of the transformer are separately connected to the diode connection point, the leakage inductance of the transformer is substantially increased. Such an increase in the leakage inductance reduces the effect of suppressing a surge voltage generated in the full-bridge converter.

(7) Therefore, an object of the present disclosure is to provide a composite component that is less likely to impair the effect of suppressing a surge voltage generated in a full-bridge converter.

SUMMARY

(8) A composite component according to the present disclosure includes a primary winding, a secondary winding, a coil, a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, and wiring. The primary winding and the secondary winding in

combination function as a transformer. The secondary winding is connected between the third terminal and the fourth terminal. The fifth terminal functions as a center tap of the secondary winding. The sixth terminal is connected to the first terminal via the wiring and the primary winding. The sixth terminal is connected to the second terminal via the wiring and the coil. A first portion of the primary winding that is connected to the wiring, a second portion of the coil that is connected to the wiring, and the wiring constitute a three-way junction.

ADVANTAGEOUS EFFECTS OF INVENTION

(9) With the composite component according to the present disclosure, the effect of suppressing a surge voltage generated in a full-bridge converter is less likely to be impaired.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a circuit diagram illustrating a full-bridge converter.
- (2) FIG. 2 is a plan view showing a transformer and an inductor in a comparative example.
- (3) FIG. 3 is a plan view showing the transformer and the inductor in the comparative example.
- (4) FIG. 4 is a circuit diagram of a full-bridge converter when the comparative example is adopted.
- (5) FIG. 5 is a perspective view illustrating a composite component according to a first embodiment.
- (6) FIG. 6 is a side view illustrating the composite component according to the first embodiment.
- (7) FIG. 7 is a perspective view showing parts of the composite component according to the first embodiment, with intervals therebetween widened along one direction.
- (8) FIG. 8 is a circuit diagram of a full-bridge converter when each of the composite components according to the first embodiment, a second embodiment, a third embodiment, and a fourth embodiment is adopted.
- (9) FIG. 9 is a perspective view illustrating the composite component according to the second embodiment.
- (10) FIG. 10 is a side view illustrating the composite component according to the second embodiment.
- (11) FIG. 11 is a perspective view showing parts of the composite component according to the second embodiment, with intervals therebetween widened along one direction.
- (12) FIG. 12 is a perspective view illustrating the composite component according to the third embodiment.
- (13) FIG. 13 is a side view illustrating the composite component according to the third embodiment.
- (14) FIG. 14 is a perspective view showing parts of the composite component according to the third embodiment, with intervals therebetween widened along one direction.
- (15) FIG. 15 is a perspective view illustrating the composite component according to the fourth embodiment.
- (16) FIG. 16 is a side view illustrating the composite component according to the fourth embodiment.
- (17) FIG. 17 is a perspective view showing parts of the composite component according to the fourth embodiment, with intervals therebetween widened along one direction.

DESCRIPTION OF EMBODIMENTS OF THE PRESENT DISCLOSURE

(18) First, aspects of the present disclosure will be listed and described. The present disclosure is as follows.

(19) A composite component includes a primary winding, a secondary winding, a coil, a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, and wiring. The primary winding and the secondary winding in combination function as a transformer.

The secondary winding is connected between the third terminal and the fourth terminal. The fifth terminal functions as a center tap of the secondary winding. The sixth terminal is connected to the first terminal via the wiring and the primary winding. The sixth terminal is connected to the second terminal via the wiring and the coil. A first portion of the primary winding that is connected to the wiring, a second portion of the coil that is connected to the wiring, and the wiring constitute a three-way junction.

(20) In a full-bridge converter, when a change occurs from a first state in which the voltage of the transformer on the primary side is very low, to a second state in which the voltage of the transformer on the primary side is high, no current flows through the wiring. Therefore, the inductance of the wiring does not affect the surge voltage. Since the inductance of the first portion is reduced, the effect of suppressing the surge voltage is less likely to be impaired.

(21) It is preferable that each of the primary winding, the secondary winding, and the coil is wound with a first direction serving as an axial direction, the primary winding and the coil are disposed adjacent to each other in a second direction, and the second direction is different from the first direction.

(22) Since the primary winding and the coil can be disposed at substantially the same position in the first direction, the composite component can be formed flat, and hence compact.

(23) It is preferable that the sixth terminal is located away from the primary winding and the coil along a third direction, the third direction is nonparallel to a plane determined by the first direction and the second direction, and each of a direction in which the primary winding is wound relative to the first direction as the primary winding extends from the first terminal toward the first portion, and a direction in which the coil is wound relative to the first direction as the coil extends from the second terminal toward the second portion is one of two rotation directions as viewed along the first direction, the rotation direction having a larger rotation angle required to reach the third direction from the second direction. The reason being that the ratio of the inductance contributing to suppression of the surge voltage is likely to be set large.

(24) It is preferable that a portion of the primary winding that includes at least the first portion, a portion of the coil that includes at least the second portion, and the wiring are formed of a monolithic steel plate. The reason being that the three-way junction can be easily formed.

(25) It is preferable that the primary winding is formed of a first flat rectangular wire, and the coil is formed of a second flat rectangular wire, and the second portion of the second flat rectangular wire is connected to the first flat rectangular wire in the first direction, with the first flat rectangular wire including the first portion and the wiring, or the first portion of the first flat rectangular wire is connected to the second flat rectangular wire in the first direction, with the second flat rectangular wire including the second portion and the wiring. The reason being that a multi-layer primary winding and a multi-layer coil can be easily formed.

(26) Specific examples of a composite component according to the present disclosure will be described below with reference to the drawings. It should be noted that the present disclosure is not limited to these examples, but is defined by the claims, and is intended to include all modifications which fall within the scope of the claims and the meaning and scope of equivalents thereof.

(27) Description of Full-Bridge Converter

(28) FIG. 1 is a circuit diagram illustrating a converter **200** that employs a transformer **100** and an inductor **3**. The converter **200** is a full-bridge converter.

(29) Terminals **11** and **14** of the transformer **100** function as primary-side terminals of the transformer **100**. A primary winding **10** of the transformer **100** is connected between the terminals **11** and **14**. Note that an inductor **La1** illustrated as being located between the terminal **11** and the primary winding **10** inside the transformer **100** equivalently indicates the leakage inductance of the transformer **100** on the primary side. In the following, for the sake of convenience, the primary winding **10** and the terminals **11** and **14** may be collectively referred to as a primary-side structure **1**.

- (30) Terminals **21a**, **21b**, and **21c** of the transformer **100** function as secondary-side terminals of the transformer **100**. A secondary winding **20** of the transformer **100** is connected between the terminal **21a** and the terminal **21b**. The terminal **21c** functions as a center tap of the secondary winding **20**. In the following, for the sake of convenience, the secondary winding **20** and the terminals **21a**, **21b**, and **21c** may be collectively referred to as a secondary-side structure **2**.
- (31) Switching elements **Q1**, **Q2**, **Q3**, and **Q4** and diodes **D1** and **D2** are provided between power lines **H1** and **L1** on the primary side of the transformer **100**. The power line **H1** has a potential that is higher than that of the power line **L1**.
- (32) The switching elements **Q1**, **Q2**, **Q3**, and **Q4** correspond to the above-described two pairs of switch elements. The diodes **D1** and **D2** correspond to the above-described pair of clamp diodes. The power line **H1** corresponds to the above-described high-voltage line. The power line **L1** corresponds to the above-described low-voltage line.
- (33) The switching elements **Q1** and **Q2** are connected in series between the power lines **H1** and **L1** via a connection point **P1**. The switching elements **Q3** and **Q4** are connected in series between the power lines **H1** and **L1** via a connection point **P2**.
- (34) The cathode of the diode **D1** is connected to the power line **H1**. The anode of the diode **D2** is connected to the power line **L1**. The diodes **D1** and **D2** are connected in series between the power lines **H1** and **L1** via a connection point **P3**. The anode of the diode **D1** and the cathode of the diode **D2** are connected to the terminal **11** via the connection point **P3**. The connection point **P3** corresponds to the above-described diode connection point. The terminal **11** can be used as the connection point **P3**.
- (35) The connection point **P3** is connected to the connection point **P1** via an inductor **3**. A terminal **32** of the inductor **3** is connected to the connection point **P1**. A terminal **31** of the inductor **3** is connected to the connection point **P3**. The terminal **14** is connected to the connection point **P2**. The terminal **14** can be used as the connection point **P2**.
- (36) Switching elements **Q101** and **Q102**, an inductor **Lc**, and a capacitor **Cd** are provided on the secondary side of the transformer **100**. The capacitor **Cd** is provided between power lines **H2** and **L2**. The power line **H2** has a potential that is higher than that of the power line **L2**. The power line **L2** is grounded, for example.
- (37) One end of the switching element **Q101** is connected to the terminal **21b**, and the other end thereof is connected to the power line **L2**. One end of the switching element **Q102** is connected to the terminal **21a**, and the other end thereof is connected to the power line **L2**. One end of the inductor **Lc** is connected to the terminal **21c**, and the other end thereof is connected to the power line **H2**.
- (38) All of the switching elements **Q1**, **Q2**, **Q3**, **Q4**, **Q101**, and **Q102** are realized by a field-effect transistor, for example.
- (39) Since the operations of the converter **200** having the above-described configuration, including, for example, the timing at which the switching elements **Q1**, **Q2**, **Q3**, **Q4**, **Q101**, and **Q102** are switched, are known, descriptions of the operations are omitted in the present embodiment. A description will be given of the benefit of reducing the leakage inductance on the primary side of the transformer **100**.
- (40) The inductor **3** has the function of reducing the surge voltage on the secondary side of the converter **200**. Energy is regenerated to the power lines **H1** and **L1** via the diodes **D1** and **D2**.
- (41) The larger the ratio of the inductance (hereinafter referred to as an “inductance **Lb**”) of the inductor **3** to the leakage inductance, the greater the effect of reducing the surge voltage on the secondary side is. The sum of the inductance **Lb** and the leakage inductance affects the resonant period of so-called soft switching. In view of this effect, it is not desirable to increase the inductance **Lb** without limitation. Therefore, it is desirable that leakage inductance is small.
- DESCRIPTION OF COMPARATIVE EXAMPLE**
- (42) FIGS. **2** and **3** are plan views showing a transformer **100** and an inductor **3** in a comparative

example. In the comparative example, the transformer **100** and the inductor **3** are provided separate from each other.

(43) The transformer **100** includes a core **5**. The primary winding **10** and the secondary winding **20** are wound around the core **5**. The axial direction of this winding is shown as the Z direction in the drawings. To keep the drawing from becoming complex, the core **5** is indicated by a dashed double-dotted line indicating its position in FIGS. **2** and **3**. The core **5** is an EI-type core, for example. To keep the drawing from becoming complex, the secondary winding **20** is not shown in FIG. **2**, and is indicated by a dashed double-dotted line indicating its position in FIG. **3**. The primary winding **10** and the secondary winding **20** are coupled to each other with the same polarity via the core **5**.

(44) The inductor **3** includes a coil **30**, terminals **31** and **32**, and a core **33**. The coil **30** is connected between the terminals **31** and **32**. The coil **30** is wound around the core **33**. The axial direction of this winding is shown as the Z direction in the drawings. To keep the drawing from becoming complex, the core **33** is indicated by a dashed double-dotted line indicating its position in FIGS. **2** and **3**. The core **33** is an EI-type core, for example.

(45) A direction extending from the transformer **100** toward the inductor **3** is shown as the X direction in the drawings. Here, a case is illustrated where the X direction is orthogonal to the Z direction. In FIGS. **2** and **3**, a case is illustrated where the Y direction is orthogonal to both the X direction and the Z direction. As viewed along the Z direction, the Y direction can be reached clockwise with an angle of less than π radian from the X direction. That is, so-called right-handed orthogonal directions are adopted for the X direction, the Y direction, and the Z direction in this order. The right-handed orthogonal directions are commonly adopted for all of the embodiments described below.

(46) A case is illustrated where the terminals **11** and **14** are disposed on the Y-direction side of the primary-side structure **1**, as viewed from the primary winding **10**. A case is illustrated where the terminals **31** and **32** are disposed on the Y-direction side of the inductor **3**, as viewed from the coil **30**.

(47) The primary winding **10** has a first portion **10s** on the terminal **11** side. The first portion **10s** is shown with dot hatching in order to indicate its position. The first portion **10s** is a portion of the primary winding **10**, but is not surrounded by the core **5**, and does not connect portions of the primary winding **10** that are surrounded by the core **5**. Accordingly, the first portion **10s** functions as wiring without being coupled to the secondary winding **20**.

(48) The coil **30** has a second portion **30s** on the terminal **31** side thereof. The second portion **30s** is shown with dot hatching in order to indicate its position. The second portion **30s** is a portion of the coil **30**, but is not surrounded by the core **33**, and does not connect portions of the coil **30** that are surrounded by the core **33**. Accordingly, the second portion **30s** functions as wiring.

(49) FIG. **4** is a circuit diagram of a full-bridge converter when the transformer **100** and the inductor **3** illustrated in FIGS. **2** and **3** are adopted. Note that portions that are not necessary for the following description are omitted, and only the transformer **100** and the inductor **3**, as well as the vicinity thereof are shown.

(50) The inductor La is shown separated into the inductors La1 and La2. Although the first portion **10s** constitutes a portion of the primary winding **10**, the inductance of the first portion **10s** is equivalently represented by the inductor La2. The inductor La2 is located in series with the inductor La1 between the terminal **11** and the primary winding **10**. Accordingly, the first portion **10s** equivalently increases the leakage inductance of the transformer **100**. The inductor La indicates the leakage inductance of the transformer **100**, taking the first portion **10s** into account.

(51) A portion that is not surrounded by the core **5**, and that does not connect portions of the primary winding **10** that are surrounded by the core **5** is present on the terminal **14** side of the primary winding **10**. This portion is not particularly related to the following description, and therefore the inductance of this portion is considered as being included in the inductor La1.

(52) Although the second portion **30s** constitutes a portion of the coil **30**, the inductance of the second portion **30s** is equivalently represented by the inductor **Lb2**. The inductor **Lb1** equivalently represents the inductance of portions of the coil **30** other than the second portion **30s**. The inductors **Lb1** and **Lb2** are connected in series between the terminals **31** and **32**. The inductance **Lb** is an inductance obtained by combining the inductors **Lb1** and **Lb2**.

(53) A portion that is not surrounded by the core **33**, and that does not connect portions of the coil **30** that are surrounded by the core **33** is present on the terminal **32** side of the coil **30**. This portion is not particularly related to the following description, and therefore the inductance of this portion is considered as being included in the inductor **Lb1**.

(54) The first portion **10s** effectively increases the leakage inductance of the transformer **100**, thus impairing the effect of reducing the surge voltage using the inductor **3**.

(55) Let us assume that a change occurs from a first state in which the voltage of the transformer **100** on the primary side is very low (0 V when the on-resistances of the switching elements **Q1**, **Q2**, **Q3**, and **Q4** are ignored) to a second state in which the voltage of the transformer **100** on the primary side is high (the voltage across the power lines **H1** and **L1** when the on-resistances of the switching elements **Q1**, **Q2**, **Q3**, and **Q4** are ignored). Due to a current flowing in the first state through the inductors **Lb1**, **Lb2**, **La1**, and **La1** that are connected in series with each other between the connection points **P1** and **P2**, a capacitance parasitic on the switching elements **Q101** and **Q102** provided on the secondary side of the transformer **100** is charged.

(56) This charging causes a surge voltage, and the larger the inductance combined in the inductors **Lb1**, **Lb2**, **La1**, and **La1** is, the more significant the surge voltage is.

First Embodiment

(57) FIG. 5 is a perspective view illustrating a composite component **400A** according to a first embodiment. FIG. 6 is a side view illustrating the composite component **400A**. The composite component **400A** includes a transformer **100** and an inductor **3**. A direction extending from the transformer **100** toward the inductor **3** is shown as the X direction in the drawings.

(58) The primary winding **10** and the secondary winding **20** are wound around the core **5**. The axial direction of this winding is shown as the Z direction in the drawings. The core **5** is an EI-type core, for example. To keep the drawing from becoming complex, the secondary winding **20** is not shown in FIG. 5, and is indicated by a dashed double-dotted line indicating its position in FIG. 6. The primary winding **10** and the secondary winding **20** are coupled to each other with the same polarity via the core **5**, and in combination function as the transformer **100**.

(59) Terminals **21a**, **21b**, and **21c** function as secondary-side terminals of the transformer **100**. The secondary winding **20** is connected between the terminal **21a** and the terminal **21b**. The terminal **21c** functions as a center tap of the secondary winding **20**.

(60) The inductor **3** includes a coil **30**, a terminal **32**, and a core **33**. The coil **30** is wound around the core **33**. The axial direction of this winding is shown as the Z direction in the drawings. The core **33** is an EI-type core, for example.

(61) FIG. 7 is a perspective view showing parts of the composite component **400A**, specifically, a structure from which the cores **5** and **33** are omitted, with intervals therebetween widened along the Z direction. Each of the two dashed dotted lines extending along the Z direction connects two portions connected in the Z direction.

(62) The primary winding **10** includes a first layer **1A** and a second layer **1B**. The first layer **1A** and the second layer **1B** are located so as to be arranged in this order along the Z direction. The first layer **1A** has a first portion **10t** and a connection portion **401**. The first layer **1A** is wound counterclockwise around the core **5** as viewed along the Z direction as the first layer **1A** extends from the first portion **10t** toward the connection portion **401**.

(63) The first portion **10t** is a portion of the primary winding **10**, but is not surrounded by the core **5**, and does not connect portions of the primary winding **10** that are surrounded by the core **5**. Accordingly, the first portion **10t** functions as wiring without being coupled to the secondary

winding **20**. The first portion **10t** forms a three-way junction **J1** together with a second portion **30t** and wiring **40a**, which will be described below. In the drawings, the boundary of the first portion **10t** is virtually shown.

(64) The second layer **1B** has a connection portion **101**, and is connected to the terminal **14** on the side opposite to the connection portion **101**. In the first embodiment, the connection between the second layer **1B** and the terminal **14** is achieved by forming the second layer **1B** and the terminal **14** as one piece. The second layer **1B** is wound counterclockwise around the core **5** as viewed along the Z direction as the second layer **1B** extends from the connection portion **101** toward the terminal **14**.

(65) The connection portion **401** is inclined or protrudes toward the Z direction relative to portions of the first layer **1A** other than the connection portion **401**. The connection portion **101** is inclined or protrudes toward the side opposite to the Z direction relative to portions of the second layer **1B** other than the connection portion **101**. The connection portion **101** of the first layer **1A** and the connection portion **401** of the second layer **1B** are connected to each other in the Z direction. The first layer **1A** and the second layer **1B** are not in contact with each other at portions other than the connection portions **101** and **401**. The primary winding **10** includes the first layer **1A** and the second layer **1B**. Accordingly, the primary winding **10** is wound counterclockwise around the core **5** as viewed along the Z direction as the primary winding **10** extends from the first portion **10t** toward the terminal **14**.

(66) The coil **30** includes a first layer **3A** and a second layer **3B**. The second layer **3B** and the first layer **3A** are located so as to be arranged in this order along the Z direction. The first layer **3A** has a second portion **30t** and a connection portion **301**. The first layer **3A** is wound clockwise around the core **33** as viewed along the Z direction as the first layer **3A** extends from the second portion **30t** toward the connection portion **301**.

(67) The second portion **30t** is a portion of the coil **30**, but is not surrounded by the core **33**, and does not connect portions of the coil **30** that are surrounded by the core **33**. Accordingly, the second portion **30t** functions as wiring. The second portion **30t** forms a three-way junction **J1** together with the first portion **10t** and wiring **40a**, which will be described below. In the drawings, the boundary of the second portion **30t** is virtually shown.

(68) The second layer **3B** has a connection portion **302**, and is connected to the terminal **32** on the side opposite to the connection portion **302**. In the first embodiment, the connection between the second layer **3B** and the terminal **32** is achieved by forming the second layer **3B** and the terminal **32** as one piece. The second layer **3B** is wound clockwise around the core **33** as viewed along the Z direction as the second layer **3B** extends from the connection portion **302** toward the terminal **32**.

(69) The connection portion **302** is inclined or protrudes toward the Z direction relative to portions of the second layer **3B** other than the connection portion **302**. The connection portion **301** is inclined or protrudes toward the side opposite to the Z direction side relative to portions of the first layer **3A** other than the connection portion **301**. The connection portion **301** of the first layer **3A** and the connection portion **302** of the second layer **3B** are connected to each other in the Z direction. The first layer **3A** and the second layer **3B** are not in contact with each other at portions other than the connection portions **301** and **302**. The coil **30** includes the first layer **3A** and the second layer **3B**. Accordingly, the coil **30** is wound clockwise around the core **33** as viewed along the Z direction as the coil **30** extends from the second portion **30t** toward the terminal **32**.

(70) The conductor **4** includes first layers **1A** and **3A**, wiring **40a**, and a terminal **41**. The wiring **40a** forms the three-way junction **J1** together with the first portion **10t** and the second portion **30t**. In FIGS. 5, 6 and 7, a case is illustrated where the wiring **40a** extends in the Y direction toward the terminal **41**, the first portion **10t** extends in the X direction toward the wiring **40a**, and the second portion **30t** extends in a direction opposite to the X direction toward the wiring **40a**. In the drawings, the boundary of the wiring **40a** is virtually shown.

(71) A case is illustrated where the terminal **14** is disposed on the Y-direction side as viewed from

the second layer **1B**. A case is illustrated where the terminal **41** is disposed on the Y-direction side as viewed from the first layers **1A** and **3A**. A case is illustrated where the terminal **32** is disposed on the Y-direction side as viewed from the second layer **3B**. A case is illustrated where the terminals **14** and **41** are disposed on the Y-direction side as viewed from the primary winding **10**. A case is illustrated where the terminals **32** and **41** are disposed on the Y-direction side as viewed from the coil **30**.

(72) All of the second layers **1B** and **3B**, and the conductor **4** can be implemented using a monolithic steel plate obtained through punching, for example. It is advantageous to adopt a monolithic steel plate for a portion of the primary winding **10** that includes at least the first portion **10t**, a portion of the coil **30** that includes at least the second portion **30t**, and the wiring **40a**, because the three-way junction **J1** can be easily formed.

(73) The connection between the connection portions **301** and **302** in the Z direction, and the connection between the connection portions **101** and **401** in the Z direction can both be realized through crimping, soldering, or welding, for example.

(74) The composite component **400A** can be described as follows. The composite component **400A** includes a primary winding **10**, a secondary winding **20**, a coil **30**, a first terminal **14**, a second terminal **32**, a third terminal **21a**, a fourth terminal **21b**, a fifth terminal **21c**, a sixth terminal **41**, and wiring **40a**. The primary winding **10** and the secondary winding **20** in combination function as a transformer **100**.

(75) The secondary winding **20** is connected between the third terminal **21a** and the fourth terminal **21b**. The fifth terminal **21c** functions as a center tap of the secondary winding **20**.

(76) The sixth terminal **41** is connected to the first terminal **14** via the wiring **40a** and the primary winding **10**. The sixth terminal **41** is connected to the second terminal **32** via the wiring **40a** and the coil **30**.

(77) A first portion **10t** of the primary winding **10** that is connected to the wiring **40a**, a second portion **30t** of the coil **30** that is connected to the wiring **40a**, and the wiring **40a** constitute a three-way junction **J1**.

(78) Each of the primary winding **10**, the secondary winding **20**, and the coil **30** is wound with the Z direction serving as the axial direction. The primary winding **10** and the coil **30** are disposed adjacent to each other in the X direction, which is different from the Z direction. By adopting such an arrangement, the primary winding **10** and the coil **30** can be disposed at substantially the same position in the Z direction. Accordingly, the composite component **400A** can be formed flat, and hence compact. In the above-described example, the X direction is orthogonal to the Z direction.

(79) FIG. **8** is a circuit diagram of a full-bridge converter when the composite component **400A** according to the first embodiment is adopted in place of the transformer **100** and the inductor **3** shown in FIG. **4**. Note that portions that are not necessary for the following description are omitted, and only the composite component **400A** and the vicinity of thereof are shown.

(80) The inductor **La1** equivalently indicates the leakage inductance of the transformer **100**. The inductor **L1t** equivalently indicates the inductance of the first portion **10t**. A portion that is not surrounded by the core **5**, and that does not connect portions of the primary winding **10** that are surrounded by the core **5** is present on the terminal **14** side of the primary winding **10**. This portion is not particularly related to the following description, and therefore the inductance of this portion is considered as being included in the inductor **La1**.

(81) The inductor **L3t** equivalently indicates the inductance of the second portion **30t**. The inductor **Lb1** equivalently represents the inductance of portions of the coil **30** other than the second portion **30t**. A portion that is not surrounded by the core **33**, and that does not connect portions of the coil **30** that are surrounded by the core **33** is present on the terminal **32** side of the coil **30**. This portion is not particularly related to the following description, and therefore the inductance of this portion is considered as being included in the inductor **Lb1**.

(82) The inductor **L40** indicates the inductance of the wiring **40a**. Since the wiring **40a**, the first

portion **10t**, and the second portion **30t** constitute the three-way junction **J1**, one end of the inductor **L40**, one end of the inductor **L3t**, and one end of the inductor **L1t** are shown connected to form a Y shape or a T shape. The configuration of such a Y-shape or T-shape connection is the same as that of a star connection in a three-phase alternating-current circuit.

(83) Since the wiring **40a** is located between the three-way junction **J1** and the terminal **41**, the other end of the inductor **L40** is shown connected to the terminal **41**. Since the first portion **10t** is located on the side of the primary winding **10** that is closer to the wiring **40a**, the other end of the inductor **L1t** is shown connected to the inductor **La1**. Since the second portion **30t** is located on the side of the coil **30** that is closer to the wiring **40a**, the other end of the inductor **L3t** is shown connected to the inductor **Lb1**.

(84) In the composite component **400A**, the inductors **Lb1**, **Lb2**, **La2**, and **La1** of the comparative example correspond to the inductors **Lb1**, **L3t**, **L1t**, and **La1**, respectively. No current flows through the inductor **L40** during a transition from the first state to the second state. The inductance of the inductor **L40** does not affect the surge voltage.

(85) In the above-described comparative example, the inductor **La2** that equivalently indicates the inductance of the first portion **10s**, and the inductor **Lb2** that equivalently indicates the inductance of the second portion **30s** both contribute to the above-described surge voltage.

(86) The first portion **10t** of the composite component **400A** is shorter than the first portion **10s** of the comparative example, and $La2 > L1t$ holds true. The second portion **30t** of the composite component **400A** is shorter than the second portion **30s** of the comparative example, and $Lb2 > L3t$ holds true. Accordingly, the inductance affecting the above-described surge voltage and the surge voltage are both smaller in the composite component **400A** than those in the comparative example. Therefore, the composite component **400A** is less likely to impair the effect of suppressing the surge voltage, as compared with the comparative example.

Second Embodiment

(87) FIG. **9** is a perspective view illustrating a composite component **400B** according to the second embodiment. FIG. **10** is a side view illustrating the composite component **400B**. The composite component **400B** includes a transformer **100** and an inductor **3**. A direction extending from the transformer **100** toward the inductor **3** is shown as the X direction in the drawing.

(88) A primary winding **10** and a secondary winding **20** are wound around a core **5**. The axial direction of this winding is shown as the Z direction in the drawings. The core **5** is an EI-type core, for example. To keep the drawing from becoming complex, the secondary winding **20** is not shown in FIG. **9**, and is indicated by a dashed double-dotted line indicating its position in FIG. **10**. The primary winding **10** and the secondary winding **20** are coupled to each other with the same polarity via the core **5**, and in combination function as the transformer **100**.

(89) Terminals **21a**, **21b**, and **21c** function as secondary-side terminals of the transformer **100**. The secondary winding **20** is connected between the terminal **21a** and the terminal **21b**. The terminal **21c** functions as a center tap of the secondary winding **20**.

(90) The inductor **3** includes a coil **30**, a terminal **32**, and a core **33**. The coil **30** is wound around the core **33**. The axial direction of this winding is shown as the Z direction in the drawings. The core **33** is an EI-type core, for example.

(91) FIG. **11** is a perspective view showing parts of the composite component **400B**, specifically, a structure from which the cores **5** and **33** are omitted, with intervals therebetween widened along the Z direction. Each of the two dashed dotted lines extending along the Z direction connects two portions connected in the Z direction.

(92) The primary winding **10** includes a first layer **1C** and a second layer **1D**. The first layer **1C** and the second layer **1D** are located so as to be arranged in this order along the Z direction. The first layer **1C** has a first portion **10w** and a connection portion **402**. The first layer **1C** is wound clockwise around the core **5** as viewed along the Z direction as the first layer **1C** extends from the first portion **10w** toward the connection portion **402**.

(93) The first portion **10w** is a portion of the primary winding **10**, but is not surrounded by the core **5**, and does not connect portions of the primary winding **10** that are surrounded by the core **5**. Accordingly, the first portion **10w** functions as wiring without being coupled to the secondary winding **20**. The first portion **10w** forms a three-way junction **J2** together with a second portion **30w** and wiring **40b**, which will be described below. The boundary of the first portion **10w** is virtually shown.

(94) The second layer **1D** has a connection portion **102**, and is connected to the terminal **14** on the side opposite to the connection portion **102**. In the second embodiment, the connection between the second layer **1D** and the terminal **14** is achieved by forming the second layer **1D** and the terminal **14** as one piece. The second layer **1D** is wound clockwise around the core **5** as viewed along the **Z** direction as the second layer **1D** extends from the connection portion **102** toward the terminal **14**.

(95) The connection portion **402** is inclined or protrudes toward the **Z** direction relative to portions of the first layer **1C** other than the connection portion **402**. The connection portion **102** is inclined or protrudes toward the side opposite to the **Z** direction relative to portions of the second layer **1D** other than the connection portion **102**. The connection portion **102** of the first layer **1C** and the connection portion **402** of the second layer **1D** are connected to each other in the **Z** direction. The first layer **1C** and the second layer **1D** are not in contact with each other at portions other than the connection portions **102** and **402**. The primary winding **10** includes the first layer **1C** and the second layer **1D**. Accordingly, the primary winding **10** is wound counterclockwise around the core **5** as viewed along the **Z** direction as the primary winding **10** extends from the first portion **10w** toward the terminal **14**.

(96) The coil **30** includes a first layer **3C** and a second layer **3D**. The second layer **3D** and the first layer **3C** are located so as to be arranged in this order along the **Z** direction. The first layer **3C** has a second portion **30w** and a connection portion **303**. The first layer **3C** is wound clockwise around the core **33** as viewed along the **Z** direction as the first layer **3C** extends from the second portion **30w** toward the connection portion **303**.

(97) The second portion **30w** is a portion of the coil **30**, but is not surrounded by the core **33**, and does not connect portions of the coil **30** that are surrounded by the core **33**. Accordingly, the second portion **30w** functions as wiring. The second portion **30w** forms the three-way junction **J2** together with the first portion **10w** and wiring **40b**, which will be described below. The boundary of the second portion **30w** is virtually shown.

(98) The second layer **3D** has a connection portion **304**, and is connected to the terminal **32** on the side opposite to the connection portion **304**. In the second embodiment, the connection between the second layer **3D** and the terminal **32** is achieved by forming the second layer **3D** and the terminal **32** as one piece. The second layer **3D** is wound clockwise around the core **33** as viewed along the **Z** direction as the second layer **3D** extends from the connection portion **304** toward the terminal **32**.

(99) The connection portion **304** is inclined or protrudes toward the **Z** direction side relative to portions of the second layer **3D** other than the connection portion **304**. The connection portion **303** is inclined or protrudes toward the side opposite to the **Z** direction side relative to portions of the first layer **3C** other than the connection portion **303**. The connection portion **303** of the first layer **3C** and the connection portion **304** of the second layer **3D** are connected to each other in the **Z** direction. The first layer **3C** and the second layer **3D** are not in contact with each other at portions other than the connection portions **303** and **304**. The coil **30** includes the first layer **3C** and the second layer **3D**. Accordingly, the coil **30** is wound clockwise around the core **33** as viewed along the **Z** direction as the coil **30** extends from the second portion **30w** toward the terminal **32**.

(100) The conductor **4** includes first layers **1C** and **3C**, wiring **40b**, and a terminal **41**. The wiring **40b** forms the three-way junction **J2** together with the first portion **10w** and the second portion **30w**. In FIGS. **9**, **10**, and **11**, a case is illustrated where the wiring **40b** extends in the **Y** direction toward the terminal **41**, the first portion **10w** extends in the **X** direction toward the wiring **40b**, and the second portion **30w** extends in a direction opposite to the **X** direction toward the wiring **40b**. In the

drawing, the boundary of the wiring **40b** is virtually shown.

(101) A case is illustrated where the terminal **14** is disposed on the Y-direction side as viewed from the second layer **1D**. A case is illustrated where the terminal **41** is disposed on the Y-direction side as viewed from the first layers **1C** and **3C**. A case is illustrated where the terminal **32** is disposed on the Y-direction side as viewed from the second layer **3D**. A case is illustrated where the terminals **14** and **41** are disposed on the Y-direction side as viewed from the primary winding **10**. A case is illustrated where the terminals **32** and **41** are disposed on the Y-direction side as viewed from the coil **30**.

(102) All of the second layers **1D** and **3D** and the conductor **4** can be implemented using a monolithic steel plate obtained through punching, for example. It is advantageous to adopt a monolithic steel plate for a portion of the primary winding **10** that includes at least the first portion **10w**, a portion of the coil **30** that includes at least the second portion **30w**, and the wiring **40b**, because the three-way junction **J2** can be easily formed. The connection between the connection portions **303** and **304** in the Z direction, and the connection between the connection portions **102** and **402** in the Z direction can both be realized through crimping, soldering, or welding, for example.

(103) The composite component **400B** can be described as follows. The composite component **400B** includes a primary winding **10**, a secondary winding **20**, a coil **30**, a first terminal **14**, a second terminal **32**, a third terminal **21a**, a fourth terminal **21b**, a fifth terminal **21c**, a sixth terminal **41**, and wiring **40b**. The primary winding **10** and the secondary winding **20** in combination function as a transformer **100**.

(104) The secondary winding **20** is connected between the third terminal **21a** and the fourth terminal **21b**. The fifth terminal **21c** functions as a center tap of the secondary winding **20**.

(105) The sixth terminal **41** is connected to the first terminal **14** via the wiring **40b** and the primary winding **10**. The sixth terminal **41** is connected to the second terminal **32** via the wiring **40b** and the coil **30**.

(106) A first portion **10w** of the primary winding **10** that is connected to the wiring **40b**, a second portion **30w** of the coil **30** that is connected to the wiring **40b**, and the wiring **40b** constitute a three-way junction **J2**.

(107) The sixth terminal **41** is located away from the primary winding **10** and the coil **30** along the Y direction. The Y direction is nonparallel to a plane determined by the Z direction and the X direction. In the above-described example, the Y direction is orthogonal to both the Z direction and the X direction.

(108) Each of the primary winding **10**, the secondary winding **20**, and the coil **30** is wound with the Z direction serving as the axial direction. The primary winding **10** and the coil **30** are disposed adjacent to each other in the X direction, which is different from the Z direction. By adopting such an arrangement, the composite component **400B** can be formed flat, and hence compact, as in the case of the composite component **400A**.

(109) In the composite component **400B**, the direction in which the primary winding **10** is wound relative to the Z direction as the primary winding **10** extends from the first terminal **14** toward the first portion **10w**, and the direction in which the coil **30** is wound as viewed along the Z direction when the coil **30** extends from the second terminal **32** toward the second portion **30w** are the same direction. This direction is one of two rotation directions as viewed along the Z direction, the rotation direction having a larger rotation angle required to reach the Y direction from the X direction. In the above-described example, this direction is counterclockwise as viewed along the Z direction.

(110) In the composite component **400A** described in the first embodiment, the direction in which the primary winding **10** is wound relative to the Z direction when the primary winding **10** extends from the first terminal **14** toward the first portion **10t** is counterclockwise as viewed along the Z direction. The direction in which the coil **30** is wound relative to the Z direction when the coil **30**

extends from the second terminal **32** toward the second portion **30t** is clockwise as viewed along the Z direction. That is, the two wound directions are different from each other.

(111) A circuit diagram of a full-bridge converter when the composite component **400B** is adopted can also be represented by FIG. **8**.

(112) The inductor **La1** equivalently indicates the leakage inductance of the transformer **100**. The inductor **L1t** equivalently indicates the inductance of the first portion **10w**. The inductance of the portion of the primary winding **10** that is not surrounded by the core **5** on the terminal **14** side, and that does not connect portions of the primary winding **10** that are surrounded by the core **5** is considered as being included in the inductor **La1**.

(113) The inductor **L3t** equivalently indicates the inductance of the second portion **30w**. The inductor **Lb1** equivalently represents the inductance of portions of the coil **30** other than the second portion **30w**. The inductance of the portion of the coil **30** that is not surrounded by the core **33** on the terminal **32** side, and that does not connect portions of the core **33** that are surrounded by the coil **30** is considered as being included in the inductor **Lb1**.

(114) The inductor **L40** indicates the inductance of the wiring **40b**. Since the wiring **40b**, the first portion **10w**, and the second portion **30w** constitute the three-way junction **J2**, one end of the inductor **L40**, one end of the inductor **L3t**, and one end of the inductor **L1t** are shown connected to form a Y shape or a T shape.

(115) As in the case of the first embodiment, the other end of the inductor **L40** is shown connected to the terminal **41**, the other end of the inductor **L1t** is shown connected to the inductor **La1**, and the other end of the inductor **L3t** is shown connected to the inductor **Lb1**.

(116) As in the case of the composite component **400A**, in the composite component **400B**, the inductors **Lb1**, **Lb2**, **La2**, and **La1** of the comparative example correspond to the inductors **Lb1**, **L3t**, **L1t**, and **La1**, respectively. No current flows through the inductor **L40** during a transition from the first state to the second state. The inductance of the inductor **L40** does not affect the surge voltage.

(117) In the above-described comparative example, the inductor **La2** that equivalently indicates the inductance of the first portion **10s**, and the inductor **Lb2** that equivalently indicates the inductance of the second portion **30s** both contribute to the above-described surge voltage.

(118) The first portion **10w** of the composite component **400B** is shorter than the first portion **10s** of the comparative example, and $La2 > L1t$ holds true. The second portion **30w** of the composite component **400B** has a length comparable to that of the second portion **30s** of the comparative example, and $Lb2 \approx L3t$ holds true. Accordingly, the inductance affecting the above-described surge voltage and the surge voltage are both smaller in the composite component **400B** than those in the comparative example. Therefore, the composite component **400B** is less likely to impair the effect of suppressing the surge voltage, as compared with the comparative example.

(119) Moreover, the ratio $(Lb1 + L3t) / (L1t + La1)$ of the inductance contributing to suppression of the surge voltage is likely to be set large, as compared with that in the composite component **400A**. This is preferable from the viewpoint of substantially improving the effect of suppressing the surge voltage using the inductor **3**.

Third Embodiment

(120) FIG. **12** is a perspective view illustrating a composite component **400C** according to a third embodiment. FIG. **13** is a side view illustrating the composite component **400C**. The composite component **400C** includes a transformer **100** and an inductor **3**. A direction extending from the transformer **100** toward the inductor **3** is shown as the X direction.

(121) A primary winding **10** and a secondary winding **20** are wound around a core **5**. The axial direction of this winding is shown as the Z direction in the drawings. The core **5** is an EI-type core, for example. To keep the drawing from becoming complex, the secondary winding **20** is not shown in FIG. **12**, and is indicated by a dashed double-dotted line indicating its position in FIG. **13**. The primary winding **10** and the secondary winding **20** are coupled to each other with the same polarity

via the core **5**, and in combination function as the transformer **100**.

(122) Terminals **21a**, **21b**, and **21c** function as secondary-side terminals of the transformer **100**. The secondary winding **20** is connected between the terminal **21a** and the terminal **21b**. The terminal **21c** functions as a center tap of the secondary winding **20**.

(123) The inductor **3** includes a coil **30**, a terminal **32**, and a core **33**. The coil **30** is wound around the core **33**. The axial direction of this winding is shown as the Z direction in the drawings. The core **33** is an EI-type core, for example.

(124) FIG. **14** is a perspective view showing parts of the composite component **400C**, specifically, a structure from which the cores **5** and **33** are omitted, with intervals therebetween widened along the Z direction. Each of the two dashed dotted lines extending along the Z direction connects two portions connected in the Z direction.

(125) The primary winding **10** is wound with the Z direction serving as the axial direction. Here, a case is illustrated where the primary winding **10** is wound in two layers along the Z direction. The primary winding **10** is connected to the terminal **41** via wiring **40c**. In the third embodiment, the connection between the primary winding **10** and the wiring **40c** is achieved by forming the primary winding **10** and the wiring **40c** as one piece.

(126) The primary winding **10** has a first portion **10p** on the wiring **40c** side thereof. The primary winding **10** is connected to the terminal **14** on the side opposite to the first portion **10p**. In the third embodiment, the connection between the primary winding **10** and the terminal **14** is achieved by forming the primary winding **10** and the terminal **14** as one piece.

(127) The primary winding **10** is wound counterclockwise around the core **5** as viewed along the Z direction as the primary winding **10** extends from the first portion **10p** toward the terminal **14**.

(128) The first portion **10p** is a portion of the primary winding **10**, but is not surrounded by the core **5**, and does not connect portions of the primary winding **10** that are surrounded by the core **5**. Accordingly, the first portion **10p** functions as wiring without being coupled to the secondary winding **20**. The first portion **10p** forms a three-way junction **J3** together with the wiring **40c** and a second portion **30p**, which will be described below. The boundary of the first portion **10p** is virtually shown.

(129) The coil **30** is wound with the Z direction serving as the axial direction. Here, a case is illustrated where the coil **30** is wound in two layers along the Z direction. The coil **30** is connected to the terminal **32**. In the third embodiment, the connection between the coil **30** and the terminal **32** is achieved by forming the coil **30** and the terminal **32** as one piece.

(130) The coil **30** has a second portion **30p** on the side opposite to the terminal **32**. An end portion of the second portion **30p** that is on the side opposite to the terminal **32** functions as a connection portion **305**.

(131) The second portion **30p** is a portion of the coil **30**, but is not surrounded by the core **33**, and does not connect portions of the coil **30** that are surrounded by the core **33**. Accordingly, the second portion **30p** functions as wiring. The second portion **30p** forms the three-way junction **J3** together with the first portion **10p** and the wiring **40c**. The boundary of the second portion **30p** is virtually shown.

(132) The coil **30** is wound clockwise around the core **33** as viewed along the Z direction as the coil **30** extends from the second portion **30p** toward the terminal **32**.

(133) The connection portion **305** is connected to the wiring **40c** side of the first portion **10p** on the side opposite to the Z direction.

(134) The primary winding **10** and the coil **30** can both be implemented using a flat rectangular wire, for example. It is preferable to adopt a flat rectangular wire for both the primary winding **10** and the coil **30**, because a multi-layer primary winding **10** and a multi-layer coil **30** can be easily formed. The connection between the connection portion **305** and the first portion **10p** in the Z direction both can be realized through crimping, soldering, or welding, for example.

(135) The composite component **400C** can be described as follows. The composite component

400C includes a primary winding **10**, a secondary winding **20**, a coil **30**, a first terminal **14**, a second terminal **32**, a third terminal **21a**, a fourth terminal **21b**, a fifth terminal **21c**, a sixth terminal **41**, and wiring **40c**. The primary winding **10** and the secondary winding **20** in combination function as a transformer **100**.

(136) The secondary winding **20** is connected between the third terminal **21a** and the fourth terminal **21b**. The fifth terminal **21c** functions as a center tap of the secondary winding **20**.

(137) The sixth terminal **41** is connected to the first terminal **14** via the wiring **40c** and the primary winding **10**. The sixth terminal **41** is connected to the second terminal **32** via the wiring **40c** and the coil **30**.

(138) A first portion **10p** of the primary winding **10** that is connected to the wiring **40c**, a second portion **30p** of the coil **30** that is connected to the wiring **40c**, and the wiring **40c** constitute a three-way junction **J3**.

(139) The sixth terminal **41** is located away from the primary winding **10** and the coil **30** along the Y direction. The Y direction is nonparallel to a plane determined by the Z direction and the X direction. In the above-described example, the Y direction is orthogonal to both the Z direction and the X direction.

(140) Each of the primary winding **10**, the secondary winding **20**, and the coil **30** is wound with the Z direction serving as the axial direction. The primary winding **10** and the coil **30** are disposed adjacent to each other in the X direction, which is different from the Z direction. Accordingly, as in the cases of the composite components **400A** and **400B**, the composite component **400C** also can be formed flat, and hence compact.

(141) Each of the primary winding **10**, the secondary winding **20**, and the coil **30** is wound with the Z direction serving as the axial direction. The primary winding **10** and the coil **30** are disposed adjacent to each other in the X direction. The X direction is different from the Z direction. In the above-described example, the X direction is orthogonal to the Z direction.

(142) In the composite component **400C**, the direction in which the primary winding **10** is wound relative to the Z direction when the primary winding **10** extends from the first terminal **14** toward the first portion **10p**, and the direction in which the coil **30** is wound as viewed along the Z direction when the coil **30** extends from the second terminal **32** toward the second portion **30p** are opposite directions.

(143) A circuit diagram of a full-bridge converter when the composite component **400C** is adopted also can be represented by FIG. **8**.

(144) The first portion **10p** of the composite component **400C** is shorter than the first portion **10s** of the comparative example, and $L_{a1} > L_{1t}$ holds true. The second portion **30p** of the composite component **400B** is shorter than the second portion **30s** of the comparative example, and $L_{b2} > L_{3t}$ holds true. Accordingly, the inductance affecting the above-described surge voltage and the surge voltage are both smaller in the composite component **400C** than those in the comparative example. Therefore, the composite component **400C** is less likely to impair the effect of suppressing the surge voltage, as compared with the comparative example.

(145) Moreover, the ratio $(L_{b1} + L_{3t}) / (L_{1t} + L_{a1})$ of the inductance contributing to suppression of the surge voltage is likely to be set large, as compared with that in the composite component **400A**. This is preferable from the viewpoint of substantially improving the effect of suppressing the surge voltage using the inductor **3**.

Fourth Embodiment

(146) FIG. **15** is a perspective view illustrating a composite component **400D** according to a fourth embodiment. FIG. **16** is a side view illustrating the composite component **400D**. The composite component **400D** includes a transformer **100** and an inductor **3**. A direction extending from the transformer **100** toward the inductor **3** is shown as the X direction.

(147) A primary winding **10** and a secondary winding **20** are wound around a core **5**. The axial direction of this winding is shown as the Z direction in the drawings. The core **5** is an EI-type core,

for example. To keep the drawing from becoming complex, the secondary winding **20** is not shown in FIG. **15**, and is indicated by a dashed double-dotted line indicating its position in FIG. **16**. The primary winding **10** and the secondary winding **20** are coupled to each other with the same polarity via the core **5**, and in combination function as the transformer **100**.

(148) Terminals **21a**, **21b**, and **21c** function as secondary-side terminals of the transformer **100**. The secondary winding **20** is connected between the terminal **21a** and the terminal **21b**. The terminal **21c** functions as a center tap of the secondary winding **20**.

(149) The inductor **3** includes a coil **30**, a terminal **32**, and a core **33**. The coil **30** is wound around the core **33**. The axial direction of this winding is shown as the Z direction in the drawings. The core **33** is an EI-type core, for example.

(150) FIG. **17** is a perspective view showing parts of the composite component **400D**, specifically, a structure from which the cores **5** and **33** are omitted, with intervals therebetween widened along the Z direction. Each of the two dashed dotted lines extending along the Z direction connects two portions connected in the Z direction.

(151) The primary winding **10** is wound with the Z direction serving as the axial direction. Here, a case is illustrated where the primary winding **10** is wound in two layers along the Z direction. The primary winding **10** is connected to the terminal **41** via wiring **40d**. In the third embodiment, the connection between the primary winding **10** and the wiring **40d** is achieved by forming the primary winding **10** and the wiring **40d** as one piece.

(152) The primary winding **10** has a first portion **10q** on the wiring **40d** side. The primary winding **10** is connected to the terminal **14** on the side opposite to the first portion **10q**. In the fourth embodiment, the connection between the primary winding **10** and the terminal **14** is achieved by forming the primary winding **10** and the terminal **14** as one piece.

(153) The primary winding **10** is wound counterclockwise around the core **5** as viewed along the Z direction as the primary winding **10** extends from the first portion **10q** toward the terminal **14**.

(154) The first portion **10q** is a portion of the primary winding **10**, but is not surrounded by the core **5**, and does not connect portions of the primary winding **10** that are surrounded by the core **5**. Accordingly, the first portion **10q** functions as wiring without being coupled to the secondary winding **20**. The first portion **10q** forms a three-way junction **J4** together with the wiring **40d** and a second portion **30q**, which will be described below. The boundary of the first portion **10q** is virtually shown.

(155) The coil **30** is wound with the Z direction serving as the axial direction. Here, a case is illustrated where the coil **30** is wound in two layers along the Z direction. The coil **30** is connected to the terminal **32**. In the fourth embodiment, the connection between the coil **30** and the terminal **32** is achieved by forming the coil **30** and the terminal **32** as one piece.

(156) The coil **30** has a second portion **30q** on the side opposite to the terminal **32**. An end portion of the second portion **30q** that is on the side opposite to the terminal **32** functions as a connection portion **306**.

(157) The second portion **30q** is a portion of the coil **30**, but is not surrounded by the core **33**, and does not connect portions of the coil **30** that are surrounded by the core **33**. Accordingly, the second portion **30q** functions as wiring. The second portion **30q** forms the three-way junction **J4** together with the first portion **10q** and the wiring **40d**. The boundary of the second portion **30q** is virtually shown.

(158) The coil **30** is wound clockwise around the core **33** as viewed along the Z direction as the coil **30** extends from the second portion **30q** toward the terminal **32**.

(159) The connection portion **306** is connected to the first portion **10q** side of the wiring **40d** on the Z direction side.

(160) The primary winding **10** and the coil **30** both can be implemented using a flat rectangular wire, for example. It is preferable to adopt a flat rectangular wire for both the primary winding **10** and the coil **30**, because a multi-layer primary winding **10** and a multi-layer coil **30** can be easily

formed. The connection between the connection portion **306** and the wiring **40d** in the Z direction can both be realized through crimping, soldering, or welding, for example.

(161) The composite component **400D** can be described as follows. The composite component **400D** includes a primary winding **10**, a secondary winding **20**, a coil **30**, a first terminal **14**, a second terminal **32**, a third terminal **21a**, a fourth terminal **21b**, a fifth terminal **21c**, a sixth terminal **41**, and wiring **40d**. The primary winding **10** and the secondary winding **20** in combination function as a transformer **100**.

(162) The secondary winding **20** is connected between the third terminal **21a** and the fourth terminal **21b**. The fifth terminal **21c** functions as a center tap of the secondary winding **20**.

(163) The sixth terminal **41** is connected to the first terminal **14** via the wiring **40d** and the primary winding **10**. The sixth terminal **41** is connected to the second terminal **32** via the wiring **40d** and the coil **30**.

(164) A first portion **10q** of the primary winding **10** that is connected to the wiring **40d**, a second portion **30q** of the coil **30** that is connected to the wiring **40d**, and the wiring **40d** constitute a three-way junction **J4**.

(165) The sixth terminal **41** is located away from the primary winding **10** and the coil **30** along the Y direction. The Y direction is nonparallel to a plane determined by the Z direction and the X direction. In the above-described example, the Y direction is orthogonal to both the Z direction and the X direction.

(166) Each of the primary winding **10**, the secondary winding **20**, and the coil **30** is wound with the Z direction serving as the axial direction. The primary winding **10** and the coil **30** are disposed adjacent to each other in the X direction, which is different from the Z direction. By adopting such an arrangement, as in the cases of the composite components **400A**, **400B**, and **400C**, the composite component **400D** can also be formed flat, and hence compact.

(167) In the composite component **400D**, the direction in which the primary winding **10** is wound relative to the Z direction as the primary winding **10** extends from the first terminal **14** toward the first portion **10q**, and the direction in which the coil **30** is wound as viewed along the Z direction when the coil **30** extends from the second terminal **32** toward the second portion **30q** are the same direction. This direction is one of two rotation directions as viewed along the Z direction, the rotation direction having a larger rotation angle required to reach the Y direction from the X direction. In the above-described example, this direction is counterclockwise as viewed along the Z direction.

(168) A circuit diagram of a full-bridge converter when the composite component **400D** is adopted can also be represented by FIG. 8.

(169) The first portion **10q** of the composite component **400D** is shorter than the first portion **10s** of the comparative example, and $L_{a1} > L_{1t}$ holds true. The second portion **30q** of the composite component **400D** has a length comparable to that of the second portion **30s** of the comparative example, and $L_{b2} \approx L_{3t}$ holds true. Accordingly, the inductance affecting the above-described surge voltage and the surge voltage are both smaller in the composite component **400D** than those in the comparative example. Therefore, the composite component **400D** is less likely to impair the effect of suppressing the surge voltage, as compared with the comparative example.

(170) Moreover, the ratio $(L_{b1} + L_{3t}) / (L_{1t} + L_{a1})$ of the inductance contributing to suppression of the surge voltage is likely to be set large, as compared with that in the composite component **400A**. This is preferable from the viewpoint of substantially improving the effect of suppressing the surge voltage using the inductor **3**.

Appendix

(171) The terminal **14** and the terminal **32** may be located on the side opposite to the transformer **100** and the inductor **3** in the Y direction. The reason being that adopting such positioning does not reduce the effect realized by the three-way junctions **J1**, **J2**, **J3**, and **J4**.

(172) The cores **5** and **33** may be fixed to each other using a nonmagnetic member. Firmly fixing

the cores 5 and 33 to each other makes the configuration of each of the composite components **400A**, **400B**, **400C**, and **400D** robust. The nonmagnetic properties of the above-described member prevent undesired coupling of magnetic flux.

(173) It should be appreciated that the configurations described in the embodiments and modifications above may be combined as appropriate as long as there are no mutual inconsistencies.

Claims

1. A composite component comprising: a primary winding, a secondary winding, a coil, a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, and a wiring, wherein the primary winding and the secondary winding in combination function as a transformer, the secondary winding is connected between the third terminal and the fourth terminal, the fifth terminal functions as a center tap of the secondary winding, the sixth terminal is connected to the first terminal via the wiring and the primary winding, the sixth terminal is connected to the second terminal via the wiring and the coil, and a first portion of the primary winding is disposed on an end of the primary winding that is opposite of the first terminal is connected to the wiring, and a second portion of the coil is disposed on an end of the coil that is opposite of the second terminal, wherein the first portion and the second portion are connected to the wiring so as to form an inductor and a three-way junction, wherein the first portion and the second portion of the wiring are disposed on the same plane.
2. The composite component according to claim 1, wherein each of the primary winding, the secondary winding, and the coil is wound with a first direction serving as an axial direction, the primary winding and the coil are disposed adjacent to each other in a second direction, and the second direction is different from the first direction.
3. The composite component according to claim 2, wherein the sixth terminal is located away from the primary winding and the coil along a third direction, the third direction is nonparallel to a plane determined by the first direction and the second direction, and each of a direction in which the primary winding is wound relative to the first direction as the primary winding extends from the first terminal toward the first portion, and a direction in which the coil is wound relative to the first direction as the coil extends from the second terminal toward the second portion is one of two rotation directions as viewed along the first direction, the rotation direction having a larger rotation angle required to reach the third direction from the second direction so as to wind in one of a clockwise and counter clockwise direction.
4. The composite component according to claim 1, wherein a portion of the primary winding that includes at least the first portion, a portion of the coil that includes at least the second portion, and the wiring are formed of a monolithic steel plate.
5. The composite component according to claim 2, wherein the primary winding is formed of a first flat rectangular wire, and the coil is formed of a second flat rectangular wire, and the second portion of the second flat rectangular wire is connected to the first flat rectangular wire in the first direction, with the first flat rectangular wire including the first portion and the wiring, or the first portion of the first flat rectangular wire is connected to the second flat rectangular wire in the first direction, with the second flat rectangular wire including the second portion and the wiring.
6. The composite component according to claim 2, wherein a portion of the primary winding that includes at least the first portion, a portion of the coil that includes at least the second portion, and the wiring are formed of a monolithic steel plate.
7. The composite component according to claim 3, wherein a portion of the primary winding that includes at least the first portion, a portion of the coil that includes at least the second portion, and the wiring are formed of a monolithic steel plate.
8. The composite component according to claim 3, wherein the primary winding is formed of a first

flat rectangular wire, and the coil is formed of a second flat rectangular wire, and the second portion of the second flat rectangular wire is connected to the first flat rectangular wire in the first direction, with the first flat rectangular wire including the first portion and the wiring, or the first portion of the first flat rectangular wire is connected to the second flat rectangular wire in the first direction, with the second flat rectangular wire including the second portion and the wiring.

9. A composite component comprising: a primary winding, a secondary winding, a coil, a first terminal, a second terminal, a third terminal, a fourth terminal, a fifth terminal, a sixth terminal, and a wiring, wherein the primary winding and the secondary winding in combination function as a transformer, the secondary winding is connected between the third terminal and the fourth terminal, the fifth terminal functions as a center tap of the secondary winding, the sixth terminal is connected to the first terminal via the wiring and the primary winding, the sixth terminal is connected to the second terminal via the wiring and the coil, and a first portion of the primary winding is disposed on an end of the primary winding that is opposite of the first terminal is connected to the wiring, and a second portion of the coil is disposed on an end of the coil that is opposite of the second terminal, wherein the first portion and the second portion are connected to the wiring so as to form an inductor and a three-way junction, wherein the first portion and the second portion are coaxial with each other and the wiring is orthogonal to both the first portion and the second portion and extends along a portion of the primary winding.
